

UHH at AVeriTeC: RAG for Fact-Checking with Real-World Claims

Chinmoy Sahoo

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1 What usually happens (before this paper):

Someone might look at a few books about dogs and see if any of them say "dogs can fly." If they find a book that says "dogs can fly," they might believe it. But sometimes the books might not be very good, or they might not have enough information.

2 What this paper is trying to do (the "novelty"):

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3 Here's what's special about their super-detective kit (RAG):

3.1 More Clues, Better Search:

Imagine there are thousands and thousands of dog books. The usual way might only look at 100 books. This paper's kit helps the detective quickly find 10,000 important dog books instead! That's a lot more clues to check. And instead of just looking at 3 important sentences, they look at 10 important sentences from those books. More sentences mean more chances to find the truth.

3.2 Smart Question Asking:

The detective kit has a special way of asking questions. One way (Retrieve-Question): It finds some interesting clues first, then it makes up good questions about those clues. Then it looks again to see which clues answer those new questions best. Another way (Question-Retrieve): It takes your original big question ("Can dogs fly?") and breaks it down into smaller, easier questions like "What do dogs usually do?" or "Do dogs have wings?". Then it tries to find clues that answer those specific little questions. This helps it get to the right answer more clearly.

3.3 A Super Brain (LLM) for the Final Decision:

After finding all the best clues and questions, the kit has a "super brain" (that's the LLM) that reads everything. The super brain is told exactly how to decide if the claim is "true," "false," "not enough information," or "some clues say yes, some say no." This super brain is much smarter than the old way of deciding.

4 Why this is cool and new:

Their new super-detective kit makes the truth-finding score much higher than the old, simpler kit. It's like going from guessing a lot to being much more accurate! So, the new part is making the RAG system much better and smarter for finding the truth about things people say online. Even though other people have tried to give detectives a "kit" before (RAG), this paper makes a much, much better kit for the very tricky job of checking if real-world things on the internet are true or not.

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